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Abstract

Paleomagnetic dating of mineralization in the Kapok MVT deposit, Lennard Shelf, Western Australia

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Abstract

The Kapok Mississippi Valley-type (MVT) Zn–Pb deposit infills dilatent zones in the Kapok fault system that cuts Upper Devonian carbonates. Paleomagnetic methods have isolated an extremely weak characteristic remanent magnetization in 89 of 98 specimens from 9 sites in mineralization and host rocks. Its pole position defines an age of 351 ± 8 Ma, concordant with a $^{238}U-^{206}Pb$ calcite age of 351 ± 15 Ma from the Kapok and Goongewa deposits and a Rb–Sr sphalerite age of 357 ± 3 Ma from the Pillara deposit. The agreement of the three independent direct dating methods implies a single MVT mineralizing event throughout the Lennard Shelf shortly after host rock deposition and its subsequent rift faulting to form the Canning Basin.
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1. Introduction

The Kapok Mine is one of several Mississippi Valley-type (MVT) zinc–lead deposits found in Upper Devonian platform carbonates of the Lennard Shelf in the Canning Basin of Western Australia (Fig. 1A). Kapok's epigenetic mineralization is found in faults cutting Frasnian (~ 370 ± 2 Ma, Haq and van Eysinga, 1998) carbonates, leading to divergent views on the age of mineralization. Most recent authors (e.g. Dörling et al., 1996; Vearncombe et al., 1996; Wallace et al., 1999) favour a genetic model that is consistent with a Rb–Sr sphalerite age of 357 ± 3 Ma from the Pillara (Blendevale) Mine about 50 km to the northwest (Christensen et al., 1996) and a $^{238}U-^{206}Pb$ age of 351 ± 15 Ma for ore-stage calcites from Kapok and the Goongewa (Twelve Mile Bore) Mine (Brannon et al., 1996). Alternatively, Tompkins et al. (1994) have suggested a younger age from petrographic relationships of ~ 325 ± 25 Ma for the MVT mineralization at Cadjebut.

The first aim of this study was to directly date Lennard Shelf MVT ores paleomagnetically to compare with the existing radiometric dates. The second aim was broader. Lennard Shelf MVT mineralization is unusual because it is the only place worldwide where two fundamentally different radiometric methods have given concordant direct dates for MVT mineralization (Leach et al., 2001). In most MVT districts, different radiometric methods have given either disparate ages or failed. Conversely, paleomagnetic methods have given a date for all MVT districts examined so far, so that it is important to test the

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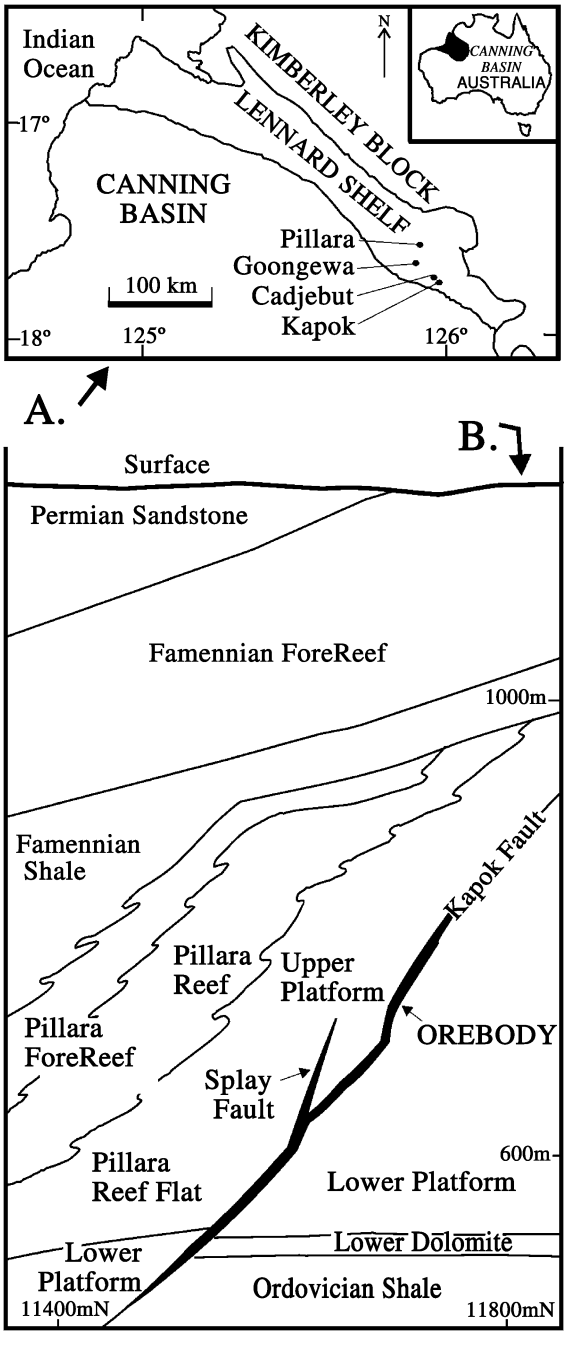


Fig. 1. (A) Location map. (B) Cross-section through the Kapok deposit.

relative utility of these methods in the Lennard Shelf deposits that have concordant radiometric ages. In 2001, paleomagnetic samples were collected at 9 sites in the Kapok Mine, reported herein, and at 23 sites from the Pillara Mine (Fig. 1) that are currently being analyzed.

2. Geology

The Lennard Shelf deposits and prospects are characterized by a diversity that spans the spectrum of MVT textural ore types. They vary from stratabound replacements at Cadjebut, to coarsely crystalline cavity fill at Goongewa, to fault-hosted deposits at Pillara and Kapok (Vearncombe et al., 1996). Despite the textural variability, the deposits are mineralogically simple, consisting of colloform to crystalline sphalerite, dendritic to crystalline galena, sparry ore-stage and post-ore calcite, colloform marcasite and, less commonly, melnikovitic pyrite.

At Kapok and Cadjebut, orthorhombic calcite psuedomorphs suggest precipitation of barite late in the paragenesis. The carbonate host rocks range in age from latest Givetian to Famennian, and were deposited in a barrier reef system (Playford, 1980) during active extension within the Canning Basin with the Precambrian Kimberley Block uplifted to the north (Fig. 1A). Field and underground observations at Pillara and Kapok indicate that mineralization occurred shortly after deposition of the Devonian host rocks during active faulting (Vearncombe et al., 1996). The Kapok Fault cuts platform limestones of the Pillara Cycle (Fig. 1B), consistently strikes southeast–northwest and generally dips between 45° and 60° to the southwest. Kinematic indicators along the fault reveal that movement was normal and predominantly dip–slip, consistent with southwest–northeast-directed extension (Kennedy, 1998). In cross-section, the fault surface varies from irregular to slightly convex, leading to an increasing amount of dilation during normal movement with increasing depth, and this correlates in general with orebody thickness.

Sulphide mineralization at Kapok is often contained between hanging wall and footwall faults, with extensive, sulphide-cemented rubble to crackle breccias in both the hanging and footwalls. It consists of repetitions of variably banded colloform sphalerite

with galena, colloform marcasite, and zoned to cloudy ore-stage calcite. This early sequence has been disrupted by continued faulting and re-cemented by crystalline sphalerite and galena. Barite may also have been precipitated during this later stage of mineralization but has been replaced by late-stage calcite. Late-stage calcite also cements post-ore breccias and suggests that fault movement continued after main-stage sulphide mineralization of the Kapok Fault had ended. Sphalerite fluid inclusion temperatures indicate peak homogenization temperatures during ore formation of ~ 90 °C (Etmian and Hoffman, 1989) and peak burial temperatures from apatite fission track measurements of 80–90 °C (Arne, 1996) at the adjacent Cadjebut MVT deposit. The Kapok fault system has hosted a total known resource of ~ 5.9 Mt at ~ 7.8% Zn, ~ 8.7% Pb, and 10–15g/t Ag.

3. Sampling and measurement

At Kapok, samples were collected from: ore-grade to massive sphalerite mineralization (sites 1, 5 and 6), marcasite mineralization (site 2), footwall limestones (sites 7 and 9), hanging wall limestones (site 4), fault breccia clasts (site 3) and a hanging wall calcite vein (site 8). The sites are spread along a 900-m fault strike length and a 400-m vertical span. Four oriented blocks were taken from each site except site 3 where eight clasts were collected. Several standard paleomagnetic specimens were drilled from each block for measurement.

Unmineralized host rock specimens weigh ~ 29.2 g, increasing to ~ 60 g as the percentage of metallic sulphides increases to ~ 100% (Table 1). The specimens' magnetic susceptibilities, measured on a Sapphire Instruments SI-2 meter, increase significantly (2σ) with mineralization (Table 1). After several weeks of storage in a magnetically shielded room (~ 100 nT) to allow viscous remanent magnetizations to decay, the specimens' natural remanent magnetizations (NRM) were measured on a 2G Enterprises vertical-axis DC-SQUID magnetometer with a sensitivity limit of $\sim 2 \times 10^{-6}$ A/m. The NRM intensities were about 1 order of magnitude weaker than most MVT deposits (Symons et al., 1996) with unmineralized and mineralized specimens giving similar median values of $\sim 2.7 \times 10^{-5}$ A/m (Table 1).

One pilot specimen per site was alternating field (AF) demagnetized in 11 steps up to 80 mT using a Sapphire Instruments SI-4 demagnetizer, reducing the remanence intensity to the magnetometer's noise level in most cases. A second pilot specimen per site was thermally demagnetized in nine steps to 330 °C using a Magnetic Measurements MM-1 demagnetizer; over ~ 330 °C their directions became erratic from either too low intensity or sulphide oxidation to magnetite. In the thermal pilot specimens from sites 1, 2, 6, 7 and 8, sharp drops of intensity were recorded between 275 and 330 °C which indicate that pyrrhotite is the major remanence carrier.

After examination of the pilot specimens' data, the approximately nine remaining specimens at each site were AF-demagnetized in eight steps up to peak values of 50–110 mT. Step demagnetization isolated specimen characteristic remanent magnetization (ChRM) directions to the north–northeast and shallowly upward or normal in most cases (~ 48.5%) or

Table 1
Lithologic properties

Lithology	Number of specimens	Weight (grams)			Susceptibility (emu)			NRM Intensity (A/m × 10^{-5})		
		Q_1	Mean	Q_3	Q_1	Mean	Q	Q_1	Mean	Q_3
Limestones	41	29.50	29.99	30.23	2.75	8.58	21.37	2.81	3.97	7.24
Fault breccia clasts	27	29.03	29.56	29.80	− 0.16	2.08	4.69	0.99	1.94	3.18
Calcitic veins	15	29.21	29.80	31.79	4.24	6.01	9.14	0.40	0.80	1.68
All host rocks	83	29.30	29.82	30.36	2.45	5.43	13.17	1.44	2.78	4.25
Sphalerite mineralization	42	35.55	37.56	47.26	9.62	12.69	20.79	1.26	1.49	4.07
Marcasite mineralization	16	30.62	31.27	34.09	90.10	115.07	149.14	12.50	19.10	24.00
All mineralizations[a]	55	33.74	36.74	42.94	10.71	15.11	32.31	1.45	2.77	5.94

[a] Specimens weighing >32 g or >10% mineralization. Q_1 and Q_3 are the 1st and 3rd quartile values. Susceptibility (χ) in emu; divide by 4π for SI equivalent.

defined remagnetization circles (Bailey and Halls, 1984) towards this direction (~ 19.6%); however, some specimens yielded antiparallel or reversed directions (~ 12.4%) or remagnetization circles to the antiparallel direction (~ 10.3%). The remaining specimens (~ 9.3%) yielded either errant or unresolvable ChRM directions. The specimen directions were combined to get the site mean and unit mean ChRM directions (Fisher, 1953; Table 2). In general, the site means represent loose clusters of specimen directions with α_{95} values between 6.5° and 15.2° and precision parameter values (k) between 11 and 63 (Table 2). Nevertheless, the resulting unit mean direction is fairly tightly clustered with $\alpha_{95}=7.7°$ and $k=45.9$, and it gives a pole position for Kapok of 63.7°S, 27.7°E (dp = 4.5°, dm = 8.3°).

Five specimens, representing several lithologies, were pulse-magnetized in 12 steps up to 900 mT using a Sapphire Instruments SI-6 DC pulse magnetizer to define their acquisition of saturation isothermal remanent magnetization (SIRM) and then the SIRM was AF-demagnetized in nine steps to 160 mT. The specimens from mineralization (sites 2 and 5), fault breccia clasts (site 3), and hanging wall limestone (site 4) have SIRM crossover values of 46 ± 5 mT, identifying pseudosingle-domain pyrrhotite as the sole significant ChRM carrier (Symons and Cioppa, 2000). One specimen from the footwall limestone (site 4) yielded an SIRM acquisition curve indicating the presence of both hematite and pyrrhotite.

Table 2
Mean remanence directions

Site	Number of specimens			Mean remanence direction			
	m	e	r	Decl. (°)	Incl. (°)	α_{95} (°)	k
1	10	5	4	25.7	−22.5	11.1	22
2	11	6	3	8.1	−34.3	11.7	20
3	15	8	5	35.1	−22.7	9.3	21
4	11	9	1	11.5	−26.8	12.2	17
5	10	4	5	41.0	−21.7	8.6	20
6	10	5	4	42.0	−16.8	6.5	63
7	10	6	4	22.1	−21.6	15.0	11
8	10	6	3	28.9	−31.5	15.2	12
9	11	10	1	22.4	−25.4	10.3	21
All nine sites				26.7	−25.2	7.7	46

Number of specimens: measured (m), giving used ChRM endpoints (e) or remagnetization circles (r). Mean remanence direction: declination (Decl.), inclination (Incl.), radius of cone of 95% confidence (α_{95}), and precision parameter (k) (Fisher, 1953).

4. Discussion

Hurley and Van der Voo (1987) first investigated the paleomagnetism of the undeformed Late Devonian reef complex along a 200 km profile of the Lennard Shelf extending northwest from Kapok. They obtained a unit mean direction of 42.7°/−28.9° (declination/inclination) with $\alpha_{95}=7.8°$ and $k=61.6$ from seven outcrop sites using data from 89 of 133 measured specimens in Upper Frasnian and Famennian red and gray limestones. Hematite is the major ChRM carrier in these specimens but goethite from recent weathering that converts to hematite at ~ 300 °C on thermal demagnetization and intense erratic magnetizations from lightning strikes were recognized as possible problems with these results. The hematite ChRM was interpreted to be syndepositional or Late Devonian in origin based on layer-parallel polarity reversals in algal mats and the equatorial paleolatitude of ~ 15°S, but the hematite need only be pre-Permian based on its being leached in fractures and around pores that were then filled with a Late Carboniferous to Early Permian burial cement. Consequently, several authors deemed this ChRM and its discordant paleopole to be a post-Devonian remagnetization (e.g. Schmidt et al., 1986, 1990; Kent and May, 1987; Li et al., 1990).

Subsequently, Chen et al. (1995) determined a primary ChRM at six outcrop sites (data used from 33 of 65 specimens) and four drill cores (19 of 32 specimens) in Frasnian and Lower Famennian strata along much the same profile with a unit mean direction of 27.5°/−19.9° with $\alpha_{95}=15.2°$ and $k=11.9$. Their results came from backreef and reefal limestones and basinal red beds, and the ChRM was shown to be primary by a syndepositional breccia test and an allochthonous block test.

The unit mean ChRM direction for the Kapok Mine is not significantly different at the 95% confidence level (2σ) (McFadden and Lowes, 1981) from that obtained by Chen et al. (1995) but is significantly different from that obtained by Hurley and Van der Voo (1987) (Fig. 2A). This indicates that the epigenetic Zn–Pb mineralizing event must have occurred

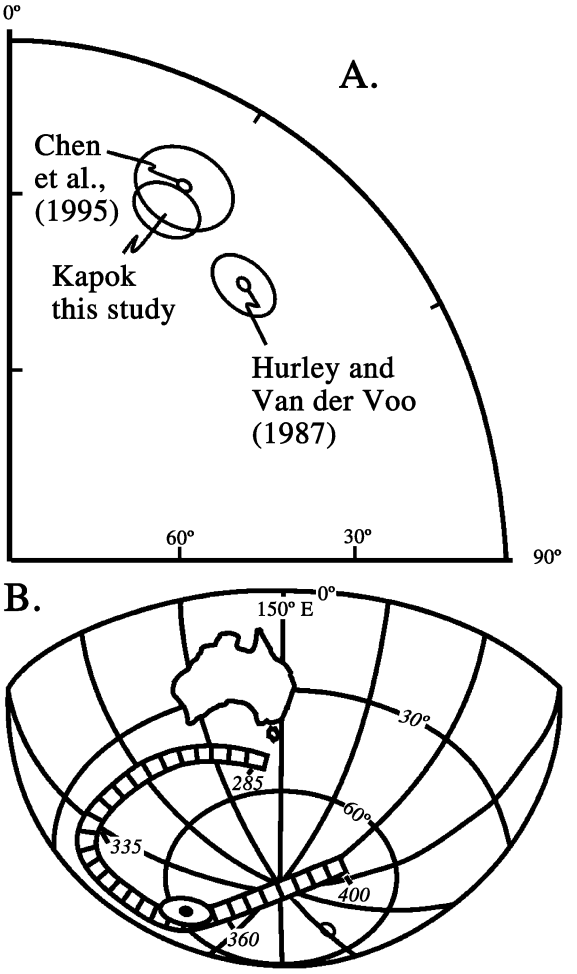


Fig. 2. (A) Equal-area stereonet, NE upper quadrant, showing unit mean ChRM directions with 95% confidence circles for Upper Devonian carbonates and mineralization in the Lennard Shelf. (B) Pole position for Kapok mineralization with its oval of 95% confidence on the apparent polar wander path of Schmidt and Clark (2000).

very shortly after deposition of the host rocks, and that it predated the hematite remagnetization event reported by the latter authors.

Recently, Schmidt and Clark (2000) have analyzed the paleomagnetic data base and generated an updated apparent polar wander path (APWP) for Australia. The pole for the MVT mineralization and host rocks at Kapok gives an age of 351 ± 8 Ma (2σ) on this path (Fig. 2B). This age is concordant with the $^{238}U-^{206}Pb$ age of 351 ± 15 Ma determined for ore-stage calcites from the Kapok and Goongewa mines by Brannon et al. (1996), providing direct confirmation of their result. The Kapok age is also concordant with the 357 ± 3 Ma Rb–Sr sphalerite age determined by Christensen et al. (1996) for a sample from the more distant Pillara Mine. This concordance suggests that there was a single MVT mineralization fluid-flow event for the Lennard Shelf during the few million years following the ~372 to 353 Ma depositional time span of the Upper Devonian limestone host rocks.

5. Conclusions

Paleomagnetism provides a direct date of 351 ± 8 Ma for the MVT Zn–Pb mineralization in the Kapok Mine, confirming the $^{238}U-^{206}Pb$ calcite age of 351 ± 15 Ma obtained by Brannon et al. (1996). The 351 ± 8 Ma age is also concordant with the 357 ± 3 Ma age determined by Christensen et al. (1996) using the Rb–Sr method on sphalerite from the Pillara Mine about 50 km away, implying contemporaneity of MVT mineralization throughout the Lennard Shelf. This issue will be further examined when results are available from the 23-site paleomagnetic collection from the Pillara Mine that is now being measured. Also, the paleomagnetic pole obtained in this study for the Kapok MVT mineralization, combined with the independent radiometric dating of it by Brannon et al. (1996), confirms the accuracy of the 351 Ma position on the APWP of Schmidt and Clark (2000).

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